

```
1  #include <WiFi.h>
2  #include <Wire.h>
3  #include <LiquidCrystal_I2C.h>
4  #include "DHT.h"
5  #include <ThingSpeak.h>
6
7  // ----- WIFI -----
8  const char* ssid = "Safii";
9  const char* password = "safiklaus";
10
11 // ----- THINGSPEAK -----
12 WiFiClient client;
13 unsigned long channelID = 3396333;
14 const char* writeAPIKey = "BxBJX5JDQ7UZU571";
15
16 // ----- WEB SERVER -----
17 WiFiServer server(80);
18
19 // ----- LCD -----
20 LiquidCrystal_I2C lcd(0x27, 16, 2);
21
22 // ----- PINS -----
23 #define DHTPIN 4
24 #define DHTTYPE DHT11
25
26 #define TRIG 5
27 #define ECHO 18
28
29 #define SOUND_PIN 34
30 #define PIEZO_PIN 35
31
32 #define BUZZER 23
33 #define LED_PIN 2
34
35 // ----- DHT -----
36 DHT dht(DHTPIN, DHTTYPE);
37
38 // ----- VARIABLES -----
39 float distance;
40 int soundValue;
41 int movementValue;
42 float temperature;
43 float humidity;
44
45 // ----- ULTRASONIC FUNCTION -----
46 float getDistance() {
47     float total = 0;
48
49     for (int i = 0; i < 5; i++) {
50         digitalWrite(TRIG, LOW);
51         delayMicroseconds(2);
52
53         digitalWrite(TRIG, HIGH);
54         delayMicroseconds(10);
55
56         digitalWrite(TRIG, LOW);
57
58         long duration = pulseIn(ECHO, HIGH, 30000);
59
60         float d = duration * 0.034 / 2;
61
62         total += d;
63
64         delay(50);
65     }
66 }
```

```
67     return total / 5;
68 }
69
70 // ----- SETUP -----
71 void setup() {
72
73     Serial.begin(115200);
74
75     pinMode(TRIG, OUTPUT);
76     pinMode(ECHO, INPUT);
77
78     pinMode(SOUND_PIN, INPUT);
79     pinMode(BUZZER, OUTPUT);
80     pinMode(LED_PIN, OUTPUT);
81
82     dht.begin();
83
84     Wire.begin();
85
86     lcd.init();
87     lcd.backlight();
88
89     lcd.setCursor(0, 0);
90     lcd.print("Sleep Monitor");
91
92     WiFi.begin(ssid, password);
93
94     lcd.setCursor(0, 1);
95     lcd.print("Connecting...");
96
97     while (WiFi.status() != WL_CONNECTED) {
98         delay(500);
99         Serial.print(".");
100    }
101
102    Serial.println();
103    Serial.println("WiFi Connected");
104
105    lcd.clear();
106    lcd.setCursor(0, 0);
107    lcd.print("WiFi Connected");
108
109    ThingSpeak.begin(client);
110
111    server.begin();
112
113    delay(2000);
114    lcd.clear();
115 }
116
117 // ----- LOOP -----
118 void loop() {
119
120     distance = getDistance();
121
122     soundValue = analogRead(SOUND_PIN);
123
124     movementValue = analogRead(PIEZO_PIN);
125
126     temperature = dht.readTemperature();
127
128     humidity = dht.readHumidity();
129
130     bool abnormal = false;
131
132     if (distance < 2 || distance > 80)
```

```
133     abnormal = true;
134
135     if (soundValue > 1500)
136         abnormal = true;
137
138     if (movementValue > 1200)
139         abnormal = true;
140
141     digitalWrite(BUZZER, abnormal);
142     digitalWrite(LED_PIN, abnormal);
143
144     lcd.clear();
145
146     lcd.setCursor(0, 0);
147     lcd.print("D:");
148     lcd.print(distance, 0);
149
150     lcd.setCursor(8, 0);
151     lcd.print("S:");
152     lcd.print(soundValue);
153
154     lcd.setCursor(0, 1);
155     lcd.print("T:");
156     lcd.print(temperature, 1);
157
158     lcd.setCursor(8, 1);
159     lcd.print("M:");
160     lcd.print(movementValue);
161
162     ThingSpeak.setField(1, distance);
163     ThingSpeak.setField(2, soundValue);
164     ThingSpeak.setField(3, movementValue);
165     ThingSpeak.setField(4, temperature);
166     ThingSpeak.setField(5, humidity);
167
168     ThingSpeak.writeFields(channelID, writeAPIKey);
169
170     WiFiClient webClient = server.available();
171
172     if (webClient) {
173
174         webClient.println("HTTP/1.1 200 OK");
175         webClient.println("Content-type:text/html");
176         webClient.println();
177
178         webClient.println("<html><head>");
179         webClient.println("<meta http-equiv='refresh' content='2'>");
180         webClient.println("<title>Sleep Monitor</title>");
181         webClient.println("</head><body>");
182
183         webClient.println("<h2>IoT Sleep Monitoring</h2>");
184         webClient.println("<p>Distance: " + String(distance) + " cm</p>");
185         webClient.println("<p>Sound: " + String(soundValue) + "</p>");
186         webClient.println("<p>Movement: " + String(movementValue) + "</p>");
187         webClient.println("<p>Temperature: " + String(temperature) + " C</p>");
188         webClient.println("<p>Humidity: " + String(humidity) + " %</p>");
189
190         if (abnormal)
191             webClient.println("<h3 style='color:red;'>ALERT DETECTED</h3>");
192         else
193             webClient.println("<h3 style='color:green;'>NORMAL CONDITION</h3>");
194
195         webClient.println("</body></html>");
196
197         webClient.stop();
198     }
```

```
199  
200     delay(2000);  
201 }  
202
```